Sketches as Previews: How Humans Sketch Visual Representations for Videos

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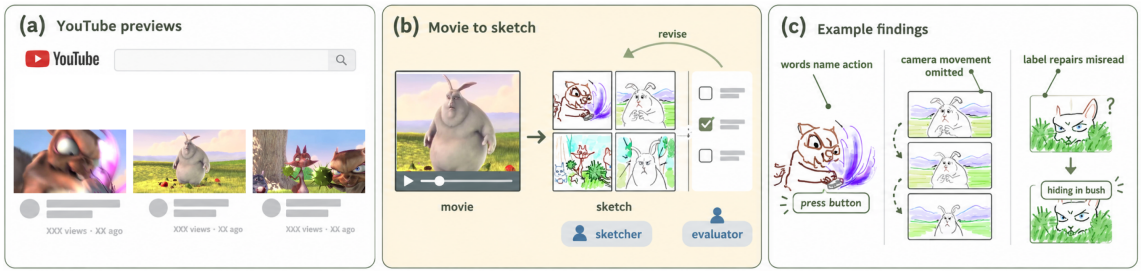


Fig. 1. (a) A conventional video preview reduces a clip to a single frame, obscuring actions and limiting source distinctiveness. (b) In the MovieDraw task, participants transformed a clip into a four-panel sketch, which peers attempted to identify; sketchers then revised the representation based on evaluation. (c) Example findings illustrate how participants compensated for the limits of static images: text was used to name actions, camera movement was often omitted, and peer misreadings prompted repairs such as explicit labels. Together, these examples show visual video captioning as a process of strategic compression in which meaning, distinctiveness, and drawability are negotiated through layered visual and textual cues.

Video previews on YouTube and Instagram has to represent an entire movie clip as static images, but thumbnails do not convey what happens, nor enable viewers to tell one source from another. To design methods of visual video captioning that convey meaning, and how feedback may improve such captions, we studied 42 people in a three-person draw-and-evaluate task to convert one movie clip into four panels of storyboard sketches. Peers then evaluated these sketches, and drawers iteratively revised the visual representations to enable recognition. We found that drawers encode subject and camera motion, character emotion, and causal relationships through channels like depictions, notations, markup text, and panel transitions. Drawers use their own perspective in evaluating others' sketches to improve their own sketches, showing the importance of the feedback process design of visual previews. We derive implications for how to design human-feedback-in-the-loop systems that meaningfully represent video content.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing**.

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1 Introduction

Video is a dominant form of online communication [20], yet because a video unfolds over time it cannot be inspected at a glance [44, 99]. Platforms therefore front videos with static visual representations: thumbnails in search and feeds, hover previews in players, and keyframe strips for navigation [34, 47, 84]. These determine what viewers see first and whether they keep watching [8, 13]. But they are *sampled* rather than constructed — chosen by first-frame heuristics,

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keyframe extraction, or low-level visual features [12, 61] — so they show what one moment looked like, not what the video is about (Figure 1a). Events and their relations [11], subject and camera motion [33], and emotion unfolding over time [103] survive only insofar as they happen to appear in the frames that were picked.

Textual captioning [1] is constructed rather than sampled, but carries its own limits. Language is lossy for composition, spatial layout, and motion character [41, 85]; a caption is bound to one language and must be reproduced for every audience a video reaches; and deployed pipelines are open-loop, offering no mechanism to improve a caption from evidence of what viewers actually recover from it. What is missing is a representation that is *visual* in form yet *constructed* to carry temporal and semantic meaning — and improvable after it is generated.

We call this *visual captioning*: the retrospective construction of a compact static visual summary of an existing video, composed of selected images optionally augmented with text, arrows, or other graphical annotation [33, 57]. It is retrospective, which distinguishes it from storyboarding, where shots are previsualized before any footage exists [2]. Visual captioning already exists in practice: schematic storyboards convert footage into annotated, arrow-marked boards [33]; video tapestries compose frames into continuous pictorial summaries [6]; comics spatialize narrative time through a conventionalized language of panels, motion lines, and balloons [14, 66]; and chronophotography flattened motion itself into single plates [10].

Every one of these, however, is the work of trained practitioners deploying learned conventions, or of systems whose designers fixed the format in advance. We know remarkably little about what people produce when no format is prescribed. This matters now because AI systems are the plausible route to visual captioning at platform scale, and the formats such systems would impose — how many frames, which annotations, what layout — currently rest on designer intuition rather than on evidence about how people encode a moving stream. Free-form drawing is an established probe of exactly this: it exposes what people select, abstract, and prioritize under communicative pressure [27, 87], and elicitation research shows that when an instrument's format misaligns with how people mentally represent the content, the resulting data are distorted or oversimplified [56].

A caption is also not a one-shot artifact. Research on graphical communication shows that when people repeatedly draw for one another and receive feedback, their representations do not simply accumulate detail but become simplified and conventionalized around what the other person could actually read [29, 31]. Revision research likewise shows that revision operations are systematic and classifiable — some target surface features, others restructure meaning — and that what revisers change reveals what they take the representation to be for [26]. If visual captions are to close the open loop of deployed captioning, systems will need to support a cycle in which captions are evaluated by viewers and revised by people, by AI, or both. Designing that cycle requires knowing which operations human revisers actually perform: whether they select new moments, redraw existing depictions, adjust only annotations, move content between image and text, or restructure the layout. These operations have not been characterized for static representations of temporal media. We therefore ask:

RQ1: How do people create meaningful visual representations of videos for the purposes of captioning?

RQ2: How do people improve these visual representations given peer feedback?

To answer these questions we ran a study with 42 participants in 14 three-person sessions, using a system we built for the purpose (Figure 1b). Each participant acted as a *sketcher* — watching a 45 s film clip, marking on a timeline the moments they intended to represent, and drawing a four-panel preview on a blank canvas — and as an *evaluator*, attempting to identify a peer's clip from that peer's sketch alone. Sketchers then revised their own work in response to what their evaluator had seen. We coded the sketches, the revisions, and replay-cued interviews, and compared each original against its revised counterpart.

What we found is that making a preview is an act of strategic compression rather than summarization: people negotiate what a moment means, what makes it distinguishable from other clips, and whether it can be drawn at all, and they layer drawing, color, arrows, and written words to carry what any one channel cannot (Figure 1c). Entities survive this translation; camera movement often does not. And because sketchers hold the source clip in mind, they cannot see which of their marks a stranger will fail to read — a gap that only the evaluator's attempt makes visible.

2 Related Work

2.1 Compact Representations of Video

Existing work represents video in two common ways. One is through text. Video captioning describes a clip, while dense event captioning describes multiple events in a longer video [1, 50, 68]. The other is through selected frames. Keyframe extraction and video summarization choose representative frames based on coverage, diversity, or interestingness [3, 35, 86, 97]. These methods decide whether the final representation should be either text or selected frames. People mainly provide annotations or evaluate the results, rather than deciding how the video should be represented.

Text descriptions are especially important when people cannot directly access the visual or audio content of a video. Audio description supports blind and low vision audiences [62, 75], while captions support deaf and hard of hearing audiences [4]. However, current captioning systems still have limitations. Captions are produced in a specific language, and much of the existing infrastructure has focused on English. This leaves multilingual audiences with less consistent coverage and quality [23, 74]. Captions are also usually generated as a finished output. They are rarely revised based on how viewers understand them. Prior work shows that feedback can improve descriptions [69, 70], and that this feedback often follows recurring patterns [71, 94]. However, prior feedback-based approaches focus on refining text rather than changing how the video is represented.

Video can also be represented visually. Most common visual summaries simply reuse frames from the original video. Thumbnails, hover previews, and keyframe strips all work in this way [12, 61]. Other systems go further by combining several moments into one new image. Pictorial summaries arrange keyframes into comic-like layouts [9, 90]. Action synopsis shows a sequence of motion in one image [5]. Schematic storyboards combine frames with outlines and arrows [33]. Tapestries, dynamic narratives, and movie-to-comic systems similarly compose multiple frames into a continuous visual summary [6, 19, 93]. These systems show that a constructed image can represent information that a single sampled frame cannot. However, the form of the visual summary is predefined by the method rather than determined by the viewer.

Static images are also widely used to communicate motion in related tasks. Storyboards help production teams communicate actions, timing, and camera changes [43, 46]. Sketch-conditioned generation uses drawings to specify how an object or scene should move [55, 81]. Recent benchmarks also test whether models can identify properties such as shot scale, composition, and camera movement from video [58, 60]. These studies examine predefined visual representations or what models can extract from video. They do not show how people would represent dynamic video content in a static image when no visual format is prescribed, or how that representation changes through feedback. Our work addresses this gap by examining how people construct their own static visual representations of video and how they revise them in response to feedback.

2.2 How People Externalize Meaning Through Drawing

Dynamic events unfold over time, whereas drawings must encode change through spatially fixed marks. Recent research identifies two complementary strategies: distributing an event across multiple panels and embedding temporal cues within a single image. In visual narratives, creators may segment motion paths into sources, routes, and goals, while using character postures, motion lines, and panel framing to communicate direction and manner. These conventions, however, are not universal. A cross-cultural corpus analysis found systematic differences in how motion paths and visual cues were distributed across panels, suggesting that motion depiction is shaped by learned representational practices [36]. A recent review further demonstrates that motion lines facilitate the comprehension of movement direction, although their interpretation draws on both perceptual correspondence and conventional visual knowledge [37].

Building on these visual conventions, storyboards extend static depictions by combining selected images with non-pictorial information. For example, Griffith, a storyboarding tool designed with Japanese animation professionals, represents scenes through illustrations, timing information, and textual descriptions [46]. This suggests that professional static representations of dynamic content function as multimodal coordination artifacts rather than as mere image sequences. Recent HCI research similarly treats arrows, paths, pose guides, force marks, and perspective cues as lightweight notations for expressing dynamic intent. Notational Animating formalizes these marks through source, path, and target representations, while showing that animator-produced notations are often contextual, composable, and ambiguous [81]. SketchDynamics further demonstrates that free-form storyboard sketches can convey motion with limited input, but may require clarification and iterative refinement to resolve underspecified intentions [55].

Together, these studies show that depicting dynamic information statically involves decisions about temporal segmentation, key states, within-frame motion cues, and textual annotation. However, recent work has primarily examined animation authoring or computational interpretation within task-specific workflows. Less is known about how people spontaneously translate existing film clips into static sketches: which moments they preserve, how they distinguish character movement from camera movement, and which visual conventions render the resulting sequence comprehensible to viewers. Our work addresses this gap by examining how people encode dynamic film information in compact sequences of sketches.

2.3 How People Depict Dynamic Information in Static Images

People can carry motion in a still picture, and a small set of devices recurs for doing so across art, science, and popular culture: off-balance posture and forward lean, blur, stroboscopic repetition, and superimposed lines and vectors [22]. Comics codified a subset of these into a shared inventory of marks [66, 91], and chronophotography arrived independently at a way of freezing a moving body into a legible sequence [10]. Cutting's survey also makes the governing constraint explicit: every workable solution evokes motion while sacrificing some accuracy of the motion represented, so solutions tend to be case-specific rather than general [21].

Most of the evidence concerns how such devices are *read*. Motion lines speed comprehension of depicted movement relative to panels without them, while reversed lines slow it, and the accompanying neural measures and comics-experience effects argue against a purely perceptual account [17]. Reviewing the competing explanations, Hacımusaoğlu and Cohn conclude that motion lines are not iconic traces of a percept but learned graphical conventions indexing conceptual path information, since their comprehension varies with development, proficiency, and culture [38]. That dependence on learning is directly measurable: motion cues differ in how effectively they convey speed, and their effect

is modulated by a reader's proficiency with manga [39]. Pictorial runes for affect have likewise been catalogued and shown to convey meaning in patterned ways [30]. This literature establishes what people *understand*, not what they would *draw*.

Where production has been studied, the referents are not video. Asked to convey routes, mechanical systems, or event orderings, people converge on a small vocabulary of schematic figures—lines, blobs, crosses, arrows—whose meaning shifts with context [87, 89]. Arrows are the clearest case: adding them shifts a mechanical diagram from a structural to a functional reading, and people spontaneously draw them when asked to convey process [42]. The devices that carry narrative time, however, are learned conventions deployed by trained practitioners. Comics convey sequence through panels, gutters, and transitions [14, 66], and corpus work shows these patterns vary systematically across traditions rather than being universal [15, 16]. Storyboard and previsualization notations are professional codes transmitted within production practice [33, 53]. The closest production evidence comes from animators: Shi et al. analyze animator-produced sketches and formalize their motion notations as source–path–target, finding them context-dependent, non-categorical, ambiguous, and composable [55, 81]—properties that make such notation expressive for the practitioner who authored it and fragile for anyone else. Their producers are professionals annotating their own intended motion for a tool to interpret; ours are untrained participants encoding an unfamiliar clip for a human stranger to identify.

We therefore know a great deal about how people *read* static depictions of motion, and about the conventions *trained* producers deploy, but little about what untrained people spontaneously produce when asked to convey a moving stream. Which devices ordinary people reinvent unaided, which they never reach for, and what they improvise instead is what separates the cognitively intuitive from what must be taught or scaffolded by an interface—precisely what a visual captioning or preview system needs to know. It is also unknown whether the devices people do produce persist, sharpen, or are replaced once a viewer's misreading is fed back to them.

2.4 How Representations Improve Through Feedback and Revision

Representations can improve as people learn what others need to understand them. In drawing games, repeated interaction produces simpler drawings and shared conventions [31]. These conventions can spread across a community as participants communicate with different partners [29]. Repetition without feedback and transmission through successive learners did not produce the same simplification [31, 32]. Simplified drawings nevertheless retain distinctive features, while their interpretation draws on shared interaction history [40]. These findings motivate alternating drawing with evaluation and examining whether revisions compress representations as well as add information.

Understanding this refinement requires distinguishing what people change from how those changes affect meaning. Writing research separates surface revisions from changes in meaning and classifies operations such as addition, deletion, substitution, and reorganization [26]. Adding information already implied by a text, for example, can preserve its meaning. Experienced writers also reconsider meaning and organization with readers in mind [83]. Later frameworks distinguish revision actions, scope, and process [18]. Applied to visual captions, these distinctions suggest examining both the edit and its target. An added label may clarify existing content, while changing an arrow may alter the represented event.

Which revisions people make also depends on the feedback they receive. Parallel prototyping and sharing alternatives improve designs and support productive critique [24, 25]. Voyant and CrowdCrit structure feedback around viewers' interpretations, visual elements, and design principles [63, 96]. CritiqueKit helps reviewers make comments specific, actionable, and justified [73]. Questions followed by statements produced stronger revisions than either format alone

[54]. However, positive reframing can increase feedback acceptance without improving revision quality [98]. Together, these studies support structured evaluation while distinguishing feedback reception, revision activity, and actual improvement.

External evaluation also offers evidence of whether descriptions communicate effectively. TurKit improved image descriptions through successive edits and crowd votes [59]. TweetTalk used clarifying conversations to identify missing information [80], while alt text research revealed differences between authors' and users' quality judgments [64]. Human and automatic feedback have also improved aspects of video descriptions [70, 72]. Beyond description tasks, peer assessments correlated with staff grades, and clearer rubrics reduced grading error [52]. Rapid peer feedback also improved final assignment performance [51]. These findings support using structured viewer judgments to guide revision and assess what others can understand.

Prior work characterizes revision in text and design, and graphical refinement in communication about individual referents. Less is known about how people revise compact, static visual representations of video. Our work addresses this gap by examining how peer feedback shapes the revision of visual captions. We characterize changes to selected moments, depictions, annotations, and the distribution of information across visual and textual channels.

3 Methods

We conducted a formative, game-based elicitation study of how people encode dynamic video content into a static visual representation, and how that representation changes once a peer attempts to identify the source from it. Participants watched a short film clip, selected the moments they intended to represent, drew on a canvas, saw a peer's attempt to identify a clip from a sketch, revised their own drawing, and then reported—per information dimension—what they had tried to convey, in which channel, and how well they believed they had succeeded. The study ran as 14 three-person sessions (42 participants). It was reviewed and approved by our institution's ethics review board.

3.1 Study rationale

Why drawing. How people internally represent information has been debated between depictive [49] and propositional [78] accounts. Without resolving that debate, drawing research has established free-form sketching as a behavioral probe of what people select, abstract, and prioritize when externalizing internal content under communicative pressure [27, 87]. We use drawing as an expressive form of visual elicitation: the externalization exposes the organization that video captioning and preview systems would need to align with.

Why moment selection is a first-class object. A video stream imposes an encoding decision with no analogue in prior drawing elicitation: choosing when. Event Segmentation Theory holds that people spontaneously parse continuous activity into discrete events at boundaries marked by change [100, 101], and that film structure interacts systematically with this parsing [65]. We therefore treat participants' selection of moments as a measure alongside the sketches themselves. Our task asked participants to select four keyframes (Section 3.4).

Why both channels. Text and visuals carry different information with different fidelity [41], and prior elicitation work found that participants distribute meaning across the two channels in construct-dependent ways [56]. We collect a drawing and a complementary written description, and additionally log on-canvas text as an object type distinct from ink strokes. This supports a per-dimension decomposition of what participants render pictorially, what they offload to language, and what they abandon.

Scope. Our primary scope is encoding: how a person turns a moving clip into a static representation, and what changes when a peer's attempt to identify it exposes a gap. Peer guessing is instrumented as a bounded decodability probe rather than a controlled receiver study, and identification in the four-alternative task was near ceiling, so we do not use guess accuracy as an outcome. Where we report representational success from self-report, we report participants' perceived success; measured communicative change is reported separately from the model-based comparison of original against revised sketches (Section 3.5.5).

3.2 Participants

We recruited 42 participants online, who formed 14 three-person sessions; a further six participants took part in piloting and are excluded from all analyses. Participants were aged 18–30 ($M = 22.38$, $SD = 2.86$); 35 identified as women and 7 as men. All sessions and all survey instruments were conducted in Chinese, and every participant reported Chinese as a language of proficiency (13 additionally reported English or another language). Participants received 30 RMB for a session of approximately 90 minutes.

Drawing experience varied but skewed toward the non-expert. On seven-point self-report scales, participants reported drawing occasionally ($Mdn = 3$, range 1–7), limited formal training in drawing, art, or design ($Mdn = 3$, range 1–7), and middling confidence in their drawing ability ($Mdn = 4$, range 1–7); they wrote to express ideas somewhat more often than they drew ($Mdn = 4$, range 1–7). Ten participants reported the minimum level of formal training and one the maximum, so the sample spans untrained to formally trained without being concentrated at either end. Full distributions are given in Appendix B.

We state the linguistic and cultural composition of the sample explicitly because visual morphology is not universal across comics traditions [16]. A codebook drawn substantially from anglophone and European comics scholarship, applied to a sample formed in a different visual-language tradition, is a scope condition on every claim we make.

A pre-session survey collected demographics, drawing and writing background, the Mini-VLAT [77], and an adaptation of the VLFI [15]. Complete Mini-VLAT totals were obtained for 33 of 42 participants ($M = 7.42$, $SD = 1.17$, range 5–9), with pronounced item-level floor and ceiling effects; we therefore use visualization literacy only as a sensitivity covariate and make no claims from it. We included the Mini-VLAT because a substantial part of our codebook consists of diagrammatic rather than depictive strategies (timeline axes, chain diagrams, operator notation), for which visualization literacy is a plausible predictor of strategy choice. The administered VLFI adaptation is not scoreable against the published instrument: the response scales differ, one item is absent, and no rule was specified for combining its two blocks. We retain the raw items and report no VLFI composite.

3.3 Stimuli

There were 12 45 s clips from Creative Commons or public-domain films, chosen to be dialogue-free or nearly so, and selected to populate a $2 \times 2 \times 2$ matrix crossing *subject motion* (high/low), *camera movement* (high/low), and *event structure* (single/multiple events). Participants are randomly assigned to one of the clips. Event boundaries and per-dimension presence were annotated independently of participants' data and are used to interpret moment selection and to validate "there wasn't much of this in the clip" responses.

3.4 Task and procedure

Sessions ran remotely over Zoom with three participants, a moderator, and three interviewers, and lasted approximately 90 minutes (about 75 minutes of task work and 15 minutes of parallel breakout interviews).

Briefing. After informed consent, participants completed the pre-session survey and were shown a demonstration of the interface. The verbal briefing described the task mechanics: watch a 45 s clip, select four keyframes that best convey its contents, and draw so that someone who had not seen the clip could understand what happens in it. Three of our five reported dimensions were therefore named to participants before they drew; cause–effect and order of events were not. We treat this as a prompting condition rather than a spontaneity condition throughout. Participants were told to work independently, that canvases were hidden from one another until share-out, and not to discuss the clip before then.

Round structure. Each participant completed ONE drawing round on one clip. Because each participant contributes one sketch-set for one clip, participant and clip are confounded, and we do not attribute differences between sketch-sets to individuals. It also follows that we cannot report the cross-round convergence analysis the protocol anticipated; interview questions about borrowing from peers are answered with respect to the revision step below rather than to a later round.

Per-round sequence. Each participant independently:

(1) **Watched** the clip (replay permitted throughout), then answered a familiarity check: "Have you seen this film before?" (yes/no) and, if yes, "How familiar are you with it?" (1–7).
(2) **Selected four keyframes** on the timeline. Prompt: *"Select the moments you plan to represent. The frames you select are references—you do not need to copy them; draw whatever best conveys the clip."*
(3) **Drew** on the canvas. Prompt: *"Create a picture, or a set of pictures, so that someone who has not watched this clip could understand what happens in it. You may draw anything and write text anywhere on the canvas."*
(4) **Saw a peer's identification attempt, then revised.** Evaluation ran in a fixed cycle within each session: the first participant evaluated the second, the second the third, and the third the first. The evaluator was shown the drawer's sketch-set without the source clip and, working only from the sketch, (i) chose which of four candidate clip descriptions it depicted, (ii) wrote a free-text account of what they believed happens in the clip, and (iii) rated the sketch-set on a 1–5 scale. The drawer was then shown the evaluator's choice, free-text account, and rating, and the pair had a brief verbal exchange. Drawers were then invited—but not required—to revise their sketch-set; 15 of 42 chose to do so. The four candidate descriptions comprised the true clip and three decoys drawn from the remaining stimulus set, sampled at random.

Breakout interviews. After the drawing task, participants entered parallel breakout rooms for an approximately 15-minute semi-structured, replay-cued walkthrough (one interviewer each, shared guide). The stroke replay was played back and participants were asked to account for their selection (why these moments; what they considered and left out; which single panel they would keep), their notation (what each non-obvious device was meant to convey; where the convention came from—comics, emoji, diagrams; which part a stranger might misread), their failures (what they wanted to express but could not draw, and what they did instead), their channel choices (why particular content went into words rather than the picture), the effect of seeing others' work, and design implications (what a tool should take over; what a tool should never do).

3.5 Analysis

Quantitative components are reported **descriptively**. Observations are nested (participants within sessions, clips within sessions), and the design is not powered for inferential claims about these factors; we report distributions, proportions,

and profiles, and declare the clustering rather than modeling it away. Where we report an interval, it is a bootstrap over the relevant clustering unit and is stated as such.

3.5.1 Moment selection. We analyze the distribution of distances from timeline marks to the nearest annotated event boundary; the relation between marked moments and each clip's annotated event count; cross-participant convergence of marked moments (proportion of participants marking within a ±2 s window); and first-stroke content, from the stroke log, as an index of encoding priority.

3.5.2 Sketch coding. Sketches were coded with a **framework-based codebook**, developed deductively from prior work on static depiction of motion [21, 38], visual morphology and narrative structure in comics [14, 66**?**], pictorial runes [30, 91], diagrammatic and arrow-based notation [42, 88], storyboard and previsualization practice [33, 53], and the drawing-elicitation codebook of Li et al. [56], and extended inductively during coding. We do not claim a purely inductive derivation: the five-theme structure was specified in advance, and codes were added, split, and merged against the corpus.

Each of the five dimensions was coded independently against five levels: **T1 Depictive** (information carried by mimetic content itself—pose, rendering, no added marks); **T2 Morphological** (conventional non-mimetic marks added on top of depiction—motion lines, arrows, runes, symbols); **T3 Structural** (information carried by relations between units—across panels, in layout, in spatial arrangement, in repetition); **T4 Textual-in-canvas** (information offloaded to written language placed on the drawing); and **T0 Absent** (the dimension is not recoverable from the drawing). Levels are not mutually exclusive: a dimension may be encoded at several levels in the same sketch-set. Each code carries a definition, inclusion and exclusion criteria, and boundary counterexamples; the full codebook appears in Table 1.

Because the four-panel format was imposed by the task, an arrangement capable of carrying order is present in every sketch-set by construction. We therefore report structural encoding of order and cause as a property of the corpus rather than as an elective strategy, and test whether that arrangement communicates order separately (§3.5.5).

Two-pass coding. Participant self-report can be used to disambiguate a drawing but must not be permitted to define what the drawing contains, so coding proceeded in two passes. In **Pass A**, coders saw only the sketch-set and the source clip and coded what was recoverable from the drawing itself. In **Pass B**, coders additionally saw the participant's localization taps, on-canvas and written text, and clarification comment, and re-coded with that privileged access. Two researchers coded independently; the first author reviewed all disagreements and produced the resolved coding used in all analyses. Each researcher coded 42 sketch-sets each coder covered and confirm that the first author's resolution is the authoritative record cited throughout.

Coders recorded a device only where it was assessable, so prevalence is reported over the sets in which a device was coded rather than over all 42; we report present, absent, and missing counts for every device.

Arrow disambiguation. Arrows recur across four dimensions (motion path, camera pan, causal link, order flow), and the same form carries different functions by context [87]. In Pass B, coders resolved function through an ordered evidence hierarchy and logged which level resolved each instance: (1) the participant's localization tap; (2) language (on-canvas text, written description, clarification) naming the relation; (3) structural position (within-panel between entities → motion or cause; between panels → order; at frame edge → camera); (4) intent tag Q1, as a weak prior; (5) coder inference, flagged as inferred. Intent tags are used as a prior, not a veto: an arrow that is structurally and linguistically unambiguous is coded as such even where the participant did not tag the corresponding dimension, since unintended-but-legible encoding is exactly what a captioning system must contend with.

3.5.3 Channel division. For each dimension we construct a funnel—*reported intent → deployment by channel → perceived success*—by crossing Q1 with coded occurrences (channel assignment disambiguated by localization taps) and Q2 confidence, capped by global adequacy. A dimension tagged as intended but absent from both channels indicates *abandoned* encoding; presence only in on-canvas text or the written description indicates *channel offloading*; deployment with low confidence indicates *perceived failure*. Q3 decomposes non-attempts into stimulus-irrelevant, capability gap, and cost, and we validate "there wasn't much of this in the clip" responses against the independent dimension-presence annotations of the stimuli.

Our drawing prompt explicitly invited on-canvas text. T4 rates and channel-offload proportions are therefore prompt-conditioned and should not be read as a natural base rate.

3.5.4 Revision coding. Revised sketch-sets were compared against their originals and classified into revision operations (adding text annotation, adding a graphical mark, adding scene content, subtracting or disambiguating, reselecting the moment, changing shot or framing, increasing depictive fidelity, refining text annotation, restructuring layout). Classifications are multi-label; a set may carry several operations. Of 18 revision records, three were excluded because the correspondence between original and revised panels could not be established

3.5.5 Model-based pairwise comparison. To test whether revision changed how recoverable a sketch-set is, we compared each original against its revised counterpart using large language models as judges. Model judges apply one criterion consistently across every pair at a scale repeated human comparison would not permit and reduce the influence of any single evaluator's preferences [102]. Following prior work we used a *pairwise* design rather than absolute scoring, which is better suited to detecting a directional change between two versions of the same artifact [48].

We used three models from different providers, none of which was involved in producing or coding the sketches: `gpt-5.6-sol` [1], `claude-sonnet-5`[2], and `gemini-3.8-flash`[3] Each with structured output enforced through the provider's schema mechanism. For each clip, the judge received the original and revised sheets together with a ground-truth summary of the clip and the four candidate descriptions among which an identification had to be made, so that identifiability was a discrimination task rather than a general quality judgment. Clip summaries and candidate descriptions were drafted from the segment records and reviewed and approved by the first author before the comparison was run. Judgments were made over the whole sketch-set, since order and causal clarity are properties of the set rather than of a single panel. Each pair was judged five times per model with the two display positions balanced across repeats (across providers the original appeared first 38 times and second 37 times), and reported estimates weight the two positions equally. This yielded **15 pairs, 225 paired model calls, and 1,350 dimension decisions**. Confidence intervals are participant-cluster bootstraps over the 15 pairs, not over individual judgments. The prompt was issued as a single user-role message containing labelled SYSTEM and USER sections rather than native role fields, and is reproduced in Appendix A. Judges evaluated six dimensions: *identifiability* (a viewer could pick the correct clip from the candidate set), *event clarity* (a viewer could say what happens, not merely what appears), *order clarity*, *causal clarity*, *motion legibility* (how subjects and the camera move), and *overall*. A residual position tendency remained after balancing (A-minus-B = −0.200, 95% CI [−0.378, −0.037]); we therefore treat this comparison as an exploratory check that corroborates the descriptive account rather than as a controlled experiment. We report the configured model

[1]OpenAI, accessed via the Responses API (`/v1/responses`). https://platform.openai.com/docs/models. Accessed 08/01/2026.
[2]Anthropic, accessed via the Messages API with header `anthropic-version: 2023-06-01`, which specifies the API protocol version rather than a model snapshot. https://docs.claude.com/en/docs/about-claude/models. Accessed 08/01/2026.
[3]Google, accessed via the Generative Language API (`v1beta`). https://ai.google.dev/gemini-api/docs/models. Accessed 08/01/2026.

identifiers; the pipeline did not retain provider-side build or snapshot identifiers, so a more specific model version cannot be recovered. All 225 calls returned valid structured output.

3.5.6 *Interviews.*

Coders and codebook development. Two researchers coded the sketch-sets. Both are HCI researchers with prior experience in visual analysis; one (the first author) had designed the study and annotated the stimuli, and the second joined at the coding stage and was not involved in data collection. The codebook was seeded deductively from the literature listed above and specified to the level of five dimensions × five encoding levels before coding began. It was then extended inductively: device-level codes within each level were added, split, and merged against the corpus, and the codebook was versioned so that every change is attributable to a specific reconciliation meeting.

Calibration and coding procedure. The two coders first independently coded a shared calibration subset of 14 sketch-sets (one third of the corpus), covering all eight cells of the stimulus matrix. They then met to compare codes line by line, surface the sources of disagreement, and revise the codebook; definitions, inclusion and exclusion criteria, and boundary counterexamples were rewritten in that meeting, and the arrow-disambiguation hierarchy described below was introduced in response to the disagreements it produced. Coding then continued against the revised codebook, with both coders independently covering 36 of the 42 sketch-sets in total and the first author additionally coding the remaining six. Segments either coder flagged as ambiguous were brought back to joint discussion rather than resolved unilaterally.

Resolution. The first author reviewed every disagreement between the two coders and produced a single resolved coding, which is the authoritative record used in all analyses and figures reported in this paper. Resolution decisions were logged with the evidence level that settled them (see the arrow hierarchy below), so that the resolved record is traceable rather than declarative.

Two-pass structure. Participant self-report can disambiguate a drawing but must not be permitted to define what the drawing contains, so both coders worked in two passes over each sketch-set. In **Pass A** they saw only the sketch-set and the source clip, and coded what was recoverable from the drawing itself. In **Pass B** they additionally saw the participant's localization taps, on-canvas and written text, and clarification comment, and re-coded with that privileged access. Pass A therefore records what a reader without the drawer's knowledge could recover, and the difference between the two passes is itself informative about the drawer's assumed common ground.

Reliability stance. Our codes are interpretive and were developed jointly through the reconciliation process above rather than applied blind to a fixed scheme, so we follow guidance for qualitative HCI work in treating a documented consensus procedure, rather than a single agreement coefficient, as the appropriate standard [67]. We report code prevalence as a description of this corpus and not as an estimate of any population quantity.

4 Results

We report how people turned a moving clip into a four-panel sketch (RQ1), and how those sketches changed after peers tried to identify the source (RQ2). A sketch is recoverable when an evaluator who has not seen the clip can identify it from the sketch alone. Each of the five content dimensions, which are subject motion, camera movement, emotion, cause–effect, and order of events, is described by its drawing level: **T0** absent, **T1** drawn directly, **T2** carried by a graphical mark such as an arrow or motion line, **T3** carried by the arrangement of panels, and **T4** written in words

(Section 3.5). Counts come from 42 coded interviews and 42 four-panel sketch sets. The sketch set, not the individual panel, is the unit throughout. The interview codebook is in Appendix 4.

4.1 Keyframe Selection Strategy for Discriminability

Nearly all participants (39 out of 42) gave an explicit rationale for what to keep, and the dominant one was *discriminability*, choosing content that separates this source from other candidates.

Rarity over recurrence (39/42). Participants favored moments that occur once over moments that repeat, because they think a repeated element cannot separate this clip from other similar clips. P23 kept a single car-flip rather than a recurring character who "appears many times … and lacks distinctiveness."

A single distinctive anchor (33/42). Participants built the set around one unusual object or configuration and expected a viewer to work outward from it. This is the same strategy participants used when identifying others' sketches (Section 4.5), which is why we report it on both sides. For example, P1 anchored their drawings on unique props such as a keychain.

Drawability as a criteria (30/42). Moments that could not be drawn clearly were discarded even when they mattered to the plot, so what reached the page was filtered by what could be drawn before it was filtered by what was important. In P1's case, they didn't draw fast motion because it is "pointless to draw".

Selection of which moments to draw then becomes strategic compression, which requires three requirements that conflict when drawing. The most meaningful action was often the hardest to draw, and the most distinctive object often identified the film without explaining what happened in it. 17 participants separated identifying the film from conveying the meaning of a moment from pinpointing the specific segment, treating these as different targets that one frame rarely serves at once: the characters' locations could "identify the shot but not the film's meaning" (P18).

4.2 Drawing strategies depend on properties of the video

When content moves from video to images representing it, the information preserved becomes highly uneven (Figure 2; Table 1).

Subject motion was never abandoned. It was absent from no sketch set, and participants encoded it through several channels at once: drawn directly in 37/42 sets, marked with arrows or motion lines in 27/42, and additionally named in words in 27/42. Rather than choosing a channel, participants stacked them.

Camera movement was left out most often. It was absent from 23/42 sets, and where present it was rarely drawn directly (4 sets), appearing instead through marks (13) or words (6). We observed camera left unencoded even in clips whose defining feature was camera movement; with one to three sketches per clip condition, we report this as an observation rather than an effect.

Emotion was carried by faces and words. It was drawn through facial expression in 25/42 sets and written in 11/42, and was left out entirely in 14/42, most often in action-heavy clips where participants prioritized what was happening over how it felt.

Order and cause were carried by the arrangement of panels. Order of events was structural in 42/42 sets and cause–effect in 40/42, while neither was drawn directly in more than one set. Because the task asked for four sequential panels, an ordering arrangement is present in every set by construction. Its presence is guaranteed; that it communicates order to a viewer is a separate question, which we return to in Section 4.7.

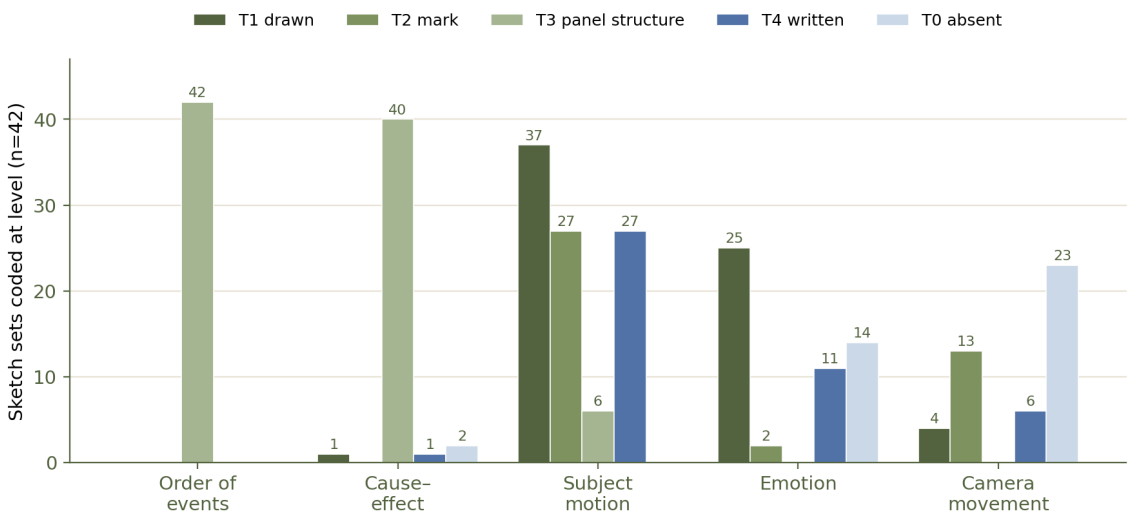


Fig. 2. How each content dimension was encoded across 42 sketch sets. Codes are multi-label, so bars within a dimension do not sum to 42. Subject motion is carried by several channels at once; camera movement is most often absent; and order and cause are carried by the arrangement of panels rather than by drawing.

4.3 Sketches fail to represent camera and motion information

Camera movement is the one dimension where drawers systematically believed they had conveyed something the page did not carry. Comparing what participants reported conveying against blind coding of their sketches, 13 of 118 dimension-cases showed no corresponding drawing method, and camera accounted for 10 of them (40% of the cases in which it was claimed). The other dimensions were near zero: subject motion 0/37, order of events 0/19, cause–effect 1/16, emotion 2/21. Comparing what participants reported conveying against blind coding of their sketches, 13 of 118 dimension-cases showed no corresponding drawing method, but these were not spread evenly. Subject motion and order of events showed no gap at all (0/37 and 0/19), emotion and cause–effect showed very few (2/21 and 1/16), and camera movement accounted for 10 of the 13 misses (40% of the cases in which it was claimed).

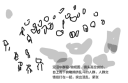
Camera is thus the one dimension where drawers systematically believed they had conveyed something the page did not carry. The interviews give the mechanism: motion is the content participants describe a frozen frame as unable to hold, and the drawing methods available for it, such as an arrow or a written note, are the ones whose meaning a drawer can read but a stranger may not. P26 observed that a mark's meaning is not guaranteed to transfer to another viewer, which was their reason for valuing the evaluation stage.

We read the residual gap as a privileged-knowledge effect: holding the clip and their own intention in mind, drawers can see a trajectory in a mark that a viewer cannot, and have no way to notice the difference. Because it is invisible to the person who produced it, it can only be surfaced by someone else (Section 4.6).

4.4 Using multimodal notation system to show movements and emotions beyond the depiction

Faced with content a literal drawing could not hold, drawers did not simply draw more carefully but they mixed pictures with notations and words (Table 1, Figure 3). Because coders recorded a drawing method only where it was assessable, we report prevalence among the sets in which each drawing method was coded.

Table 1. Drawing codebook. Drawing methods participants used to encode a clip in a sketch set, grouped by function. Coders recorded a drawing method only where it was assessable, so we report present, absent, and missing counts and the prevalence among observed sets. Mood shading is reported descriptively only.

Theme	Code	Description	Examples	Pres.	Abs.	Miss.	% obs.
Color	Color: identity/role	Distinct colors separate characters or objects		25	5	12	83.3
	Color: salience	Warm or saturated color marks the focal element		30	5	7	85.7
	Mood shading	Greyscale shading conveys atmosphere or de-emphasises background		25	2	15	92.6
Motion	Motion arrow	Arrow indicates subject movement or camera direction		22	7	13	75.9
	Motion lines	Streaks or repeated contours imply speed		7	9	26	43.8
Words	Text label	Written words name an action, sound, state, or scene		29	5	8	85.3
	Speech bubble	Quoted dialogue attached to a figure		6	10	26	37.5
	Designed typography	Stylised lettering carries emotion or emphasis		1	9	32	10.0
Reduc.	Subtraction	Deliberate omission of detail; negative space		6	4	32	60.0

Color did identity work and salience work (identity 25/30 sets; salience 30/35). Distinct colors separated characters, and a warm or saturated color marked the element a viewer was meant to look at first.

Marks carried direction (arrows 22/29). Arrows indicated where a subject or the camera moved; motion lines were less common (7/16).

Words named what a drawing could not (text labels 29/34). Words turned continuous movement into a nameable action, as one participant wrote "conducting" rather than attempt hands moving up and down, "because from a single frozen frame you don't know what the hand is doing" (P11). They also supplied depth, gaze direction and darkness,

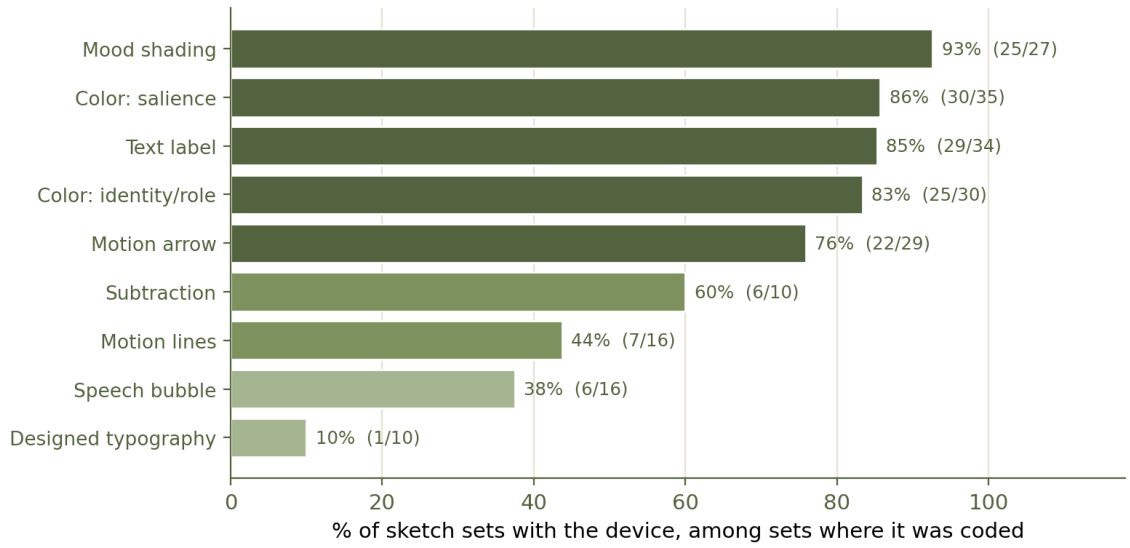


Fig. 3. Drawing methods participants used, as a percentage of the sketch sets in which each drawing method was coded. Counts are given as present/observed; observed denominators differ by drawing method.

pre-empted misreadings the drawer anticipated, and in some cases were what made identification possible at all: an evaluator recovered one clip "through the text...without it you couldn't guess" (P4). Participants further chose words for efficiency, judging a label faster to produce and more direct than an equivalent drawing.

Drawing nonetheless remained the medium participants preferred and trusted, including among those who relied most on words, and reliance on text was not a consequence of limited skill: one trained illustrator worked text-first because it offered better value for the effort, while several self-described non-drawers worked picture-first. Picture-first and text-first were stable dispositions across a participant's work rather than symptoms of ability.

4.5 Evaluators prioritize different cues than drawers

When participants took the evaluator's role, the cues they used diverged from what they had prioritized as drawers.

Spatial composition came first (25/41). Evaluators worked from the layout of the panel—where figures stood, depth, camera angle—before content: "the spatial relationship is the primary clue" (P26).

Anchor, then elimination (33/41). Evaluators locked onto one unambiguous object and used it to rule candidates out, passing over panels they judged unrepresentative.

Timeline expectations (24/41). Evaluators expected the first panel to correspond to the opening of the clip and checked whether panels were spaced plausibly across its duration.

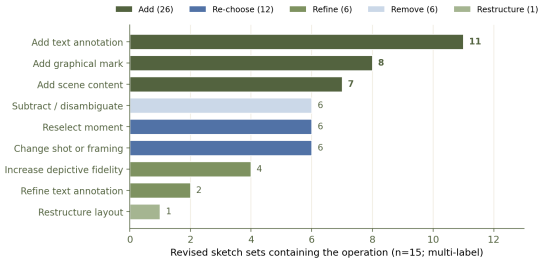
Drawers, by contrast, had spent their effort on meaning and on the most significant action. The mismatch explains why a sketch could be carefully made and still fail: effort went into the content the drawer considered important, while identification ran through composition and a single anchor.

4.6 Peer evaluation turns an invisible gap into an actionable repair

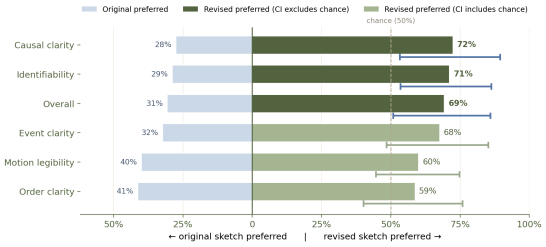
Peer evaluation gave drawers information they could not generate alone: which mark was misread, what a stranger noticed first, and which drawing methods made content recoverable.

Revisions were dominated by adding information, and most often by adding words (Figure 4a). Among the 15 sketch pairs with reliable before/after classification, adding a text annotation was the most frequent operation (11 sets), followed by adding a graphical mark (8) and adding scene content (7). Subtracting or disambiguating content, reselecting the moment, and changing the shot or framing each occurred in 6 sets; restructuring the layout occurred once. Two routes recurred in the interviews: participants discovered drawing methods by **evaluating others' sketches** (33/41), adopting text, arrows, color, or simplification after seeing a peer's work and often regretting not using them sooner; and they made **targeted repairs** when an evaluator named a specific misreading (27/41), fixing the misread element rather than redrawing the set (Figure 5).

The most common repair was thus the channel drawers had treated as secondary: feedback pushed sketches toward explicitness rather than toward greater pictorial fidelity.



(a) Revision operations after peer feedback among the same 15 sketch pairs used in Figure 4b. Operations are multi-label; a set may contribute to several bars. Bars are colored by operation type, with class totals in the legend: adding information accounts for 26 of the 51 coded operation instances, more than twice any other class.



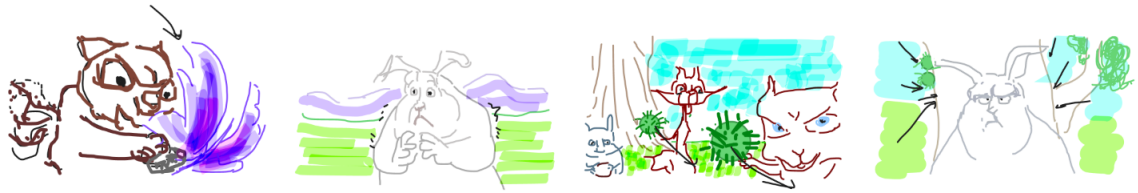
(b) Original against revised sketches, judged pairwise by three model families (15 pairs; 1,350 dimension decisions). Bars extend left for comparisons won by the original and right for those won by the revised version; whiskers are participant-cluster bootstrap intervals on the revised share. Dark bars mark intervals excluding chance.

Fig. 4. Quantitative evaluations of sketch revisions: (a) distribution of revision operations executed following feedback, and (b) pairwise AI preference win rates comparing revised against original sketches across structural dimensions.

4.7 Revised sketches are easier to identify, but no easier to follow

The revision helped identification and causal reasoning. Judges chose the revised sketch over the original in 72% of comparisons for causal clarity (95% CI [0.53, 0.89]), 71% for identifiability ([0.54, 0.86]), and 69% for overall recoverability ([0.51, 0.86]); these three intervals exclude chance (Figure 4b, Table 2).

It did not help order or motion. For event clarity the revised version won 68% of comparisons ([0.49, 0.85]), for motion legibility 60% ([0.45, 0.75]), and for order clarity 59% ([0.40, 0.76]); all three intervals include chance, so we do not claim an improvement on those dimensions. The contrast is informative rather than merely partial. The two dimensions that did not improve are the two the sketches were least able to carry to begin with: motion, which drawers most often failed to convey (Section 4.3), and order, which every set already encodes structurally by construction and for which no obvious repair exists, since adding an arrow or a further panel does not make a sequence legible if the



(a) **Color and Motion Notation**: An animated sequence relying on color coding, direction arrows, and expressive character poses to convey action and mood.



(b) **Textual and Structural Annotation**: A monochrome story sequence using text labels, dialogue callouts, and icon markers to encode dialogue and narrative context.

Fig. 5. Two qualitative sketch sequences illustrating distinct encodings: (a) visual-first representations using color and directional motion indicators, versus (b) narrative-driven line drawings combining text annotations, speech prompts, and symbolic icons.

Table 2. Position-balanced preference for the revised sketch over the original. Intervals are participant-cluster bootstraps over 15 pairs.

Dimension	Win rate	95% CI
Causal clarity	0.724	[0.533, 0.894]
Identifiability	0.711	[0.535, 0.863]
Overall	0.693	[0.509, 0.859]
Event clarity	0.676	[0.485, 0.852]
Motion legibility	0.600	[0.446, 0.748]
Order clarity	0.587	[0.402, 0.759]

grid already implied one. Revision made sketches easier to *identify* and to reason about *causally* while leaving temporal and dynamic content roughly where it was.

A residual position tendency remained after balancing (A-minus-B = −0.200, 95% CI [−0.378, −0.037]), so we treat this comparison as an exploratory check that corroborates the descriptive account rather than as a controlled experiment; it also cannot identify which revision operation produced the improvement.

4.8 What the three sources agree on

What participants said, what their sketches contained, and how the revised sketches compared converge on one dimension and diverge instructively on another.

They **agree about camera and motion**. Participants described motion as the content a frozen frame cannot hold; camera was the dimension most often absent from the sketches (23/42) and the one drawers most often believed they had conveyed when they had not (10 of 13 cases); and it is one of the two dimensions on which revision did not reliably

help (motion legibility 0.60, interval including chance). The difficulty is therefore visible in what people say, in what they draw, and in what survives revision.

They **diverge about order**, and the divergence is itself the finding. By the coding, ordering is universal: every set arranges panels in sequence, and no participant claimed to convey order without a codable device. By the model comparison, order clarity is the weakest dimension. The two are reconciled by the format: the four-panel layout guarantees that an ordering *device* is present, which satisfies a coder, while leaving open whether a viewer can recover the *sequence*, which is what the judges assessed. Presence of a structure is not the same as communication of what it structures, a distinction that matters directly for systems that assemble multi-frame previews and treat the arrangement itself as conveying time.

5 Discussion

5.1 The visual key as referential communication under privileged ground

Our central reframing that people build a *differentiating* preview rather than a depiction places this task within referential communication: designing a representation so that a specific addressee can single out a specific referent from a set of alternatives [28, 29, 31]. The closest precedent is Fan et al.'s drawing-based reference game [28], in which a sketcher draws one of four visible objects so a viewer can select it. Their participants drew more sparsely when the distractors were from different categories and more elaborately when they were close, showing that people calibrate a drawing to the discriminative work it has to do rather than to the object itself. Our drawers did the same thing one level up: they selected *moments* for rarity and distinctiveness (Section 4.1) rather than for importance, which is the temporal analogue of Fan et al.'s close-versus-far manipulation. But two differences matter. Their referent is a single static object that the viewer can see in the array; ours is a four-second event that the evaluator has never seen and that unfolds in time, so subject motion, camera movement, order, and causality must be re-encoded spatially or abandoned. And in their task the sketcher and viewer share the candidate array, whereas our drawers hold something the evaluator does not: the clip itself.

That asymmetry is the mechanism behind our central finding, and it has a direct precedent. Garrod et al. [31] found that graphical signs which two interacting partners had refined into an efficient shared system were substantially harder for non-participating observers to interpret, because meaning had migrated from the sign itself into the pair's memory of how it had been used. Our drawers were in the first half of that process without the second: having watched the clip, they read a trajectory into an arrow or a sequence into a panel grid that a stranger cannot, and—as Section 4.3 shows for camera movement—cannot tell which of those readings actually made it onto the page. The intent–uptake gap is therefore not "some content is hard to draw"; it is a failure of common ground made measurable, and it explains why peer evaluation is not an optional quality check but the only instrument that can surface it.

This lens also sharpens what is missing from how previews are made today. Thumbnail and summarization pipelines select frames by interval sampling or first-frame heuristics [3, 12, 86] and rank candidates by predicted click-through and aesthetic quality [34, 61, 84]. Song et al. [84]'s system is explicit about this: it scores frames on visual quality and aesthetic metrics and uses clustering only to check that a frame is *representative of the video*, that is, typical of its visual content. Our drawers optimized the opposite quantity. They rejected the recurring, typical character precisely because it could not separate one clip from another, and kept the once-only event. Typicality makes a frame representative; rarity makes it recoverable, and no stage of these pipelines asks whether a viewer could identify the source from the

frame chosen. Our contribution is not another selection heuristic but an account of what that missing objective is, and of how people pursue it.

5.2 Routing feedback authority by provenance

Our sharpest encoder–decoder mismatch is a design decision in disguise. Encoders optimize what a moment means and which action is salient; decoders recover clips through spatial composition—layout, position, depth, camera angle (25/41; p26: "the spatial relationship is the primary clue")—together with a shared keyframe prior, temporal-distribution matching, and locking onto a single distinctive anchor before eliminating decoys (33/41). When author intent and audience recognizability diverge, whose criteria should a feedback-driven system follow? We propose routing authority by *provenance*. For an archival or third-party clip, weight the audience/evaluator criteria, since the goal is that a viewer who does not know the clip can recognize and differentiate it, so decode-side priorities should govern. For the author's own footage, weight the author's criteria, since authorial intent about meaning can legitimately override generic recognizability. Operationally this argues for peer-review-in-the-loop authoring in which a differentiation test surfaces named misreads and the system treats them as *binding* (archival) or *advisory* (authored). And because, in our data, the value of feedback was never the score but the adaptation it triggered—the most prevalent improvement mechanism was affordance discovery *during evaluation* of peers (33/41; p2: "I should have added text and colour sooner")—a feedback system's job is to *manufacture that adaptation reliably*, not merely to rate.

5.3 Tensions we do not resolve

Four tensions in our data are genuine design conflicts rather than gaps to be trained away. We take a position on each, but none of the four dissolves.

Recognizability, meaning, and segment are not inter-tradable. A sketch can identify the film, convey the meaning of a moment, or pin the specific segment, and these pull apart: a standing position "identifies the shot but not the film's meaning" (P18). The video-summarization literature has largely not had to confront this split, because it operationalizes what-to-keep as a single quantity. Song et al.'s TVSum [97] makes the assumption unusually explicit: it treats a video's *title* as maximally descriptive of its main topic and uses title-based image search as a proxy for which shots matter, so importance is defined as *typicality with respect to the topic*. Gygli et al. [35] define it instead as visual interestingness, and Truong and Venkatesh's survey [86] organizes the field around these representativeness-style criteria. Our drawers pursued the opposite quantity. They discarded the recurring, topic-typical character precisely because it could not separate one clip from another, and kept the once-only event. Typicality makes a frame representative of a video; distinctiveness makes it recoverable to a particular segment; neither guarantees that a viewer learns what happened. Because no single frame reliably satisfies all three, authoring tools should let an author *declare which target matters* for a given preview rather than assume that a representative frame is also a recoverable or a meaningful one.

Composition versus salience. Drawers spent their effort on meaning and n the most significant action; evaluators read layout, position, depth and camera angle first (Section 4.5). This is not evaluator error. Smith's attentional theory of cinematic continuity [82] argues that continuity editing works by managing where viewers attend across cuts, so that expectations are matched compositionally rather than through a coherent internal model of the depicted space; framing, screen position and shot relations are the machinery through which film guides attention in the first place, and Block [7] treats exactly these components as the structural vocabulary of visual storytelling. Our evaluators were therefore reading film the way film is built to be read, while our drawers were encoding what the scene *meant* to them. Tversky and Heiser's finding that the same diagram is read structurally without arrows and functionally with them [42]

makes the practical point: a representation can support either reading depending on the devices present, so a system should let both be expressed rather than forcing a choice between composition and salience.

Whose criteria should govern? When author intent and audience recognizability diverge, a feedback-driven system must decide which leads. We propose routing authority by *provenance*: for an archival or third-party clip, weight the evaluator's criteria, since the goal is that someone who does not know the clip can identify it; for an author's own footage, weight the author's, since intent about meaning can legitimately override generic recognizability. The premise that a reviewer's position changes the feedback is not speculative. Natalie et al. [71] content-analyzed 1,120 comments given to novice audio-description authors and found that blind and sighted reviewers attended to systematically different qualities: the blind reviewer commented on clarity an order of magnitude more often than the sighted reviewer, and on how descriptions would sound when synthesized, while the sighted reviewer never commented on interest at all. Which reviewer you route to is therefore a substantive design choice about what kind of preview you get, not a procedural one. What our own data settles is the complementary point: the value of feedback lay in the adaptation it triggered rather than the score it produced, since the most prevalent improvement mechanism was discovering devices *while evaluating* peers (33/41). This matches what makes design critique effective elsewhere. CrowdCrit [63] scaffolds non-expert critique to be conceptual, specific and actionable, and found that designers used the resulting critiques to change their designs; PeerStudio [51] found that shortening feedback turnaround is what increases revision and subsequent performance. A feedback system's job is to manufacture that adaptation reliably, not merely to rate.

Stable individual differences. Participants differed in durable ways a one-size tool would erase. A picture-first or text-first disposition was not a skill deficit: a trained illustrator worked text-first because it offered better value for the effort, while several self-described non-drawers worked picture-first. Walny et al. [92] report the same heterogeneity in data sketching, where people across expertise levels produced representations along a continuum from literal numeracy to abstraction, with no single correct point on it. The repertoire itself is also not evenly held: comprehension of sequential images depends on fluency acquired through exposure to a graphic system rather than being universal [14], and sensitivity to motion lines varies with proficiency in reading comics [37]. Tools should therefore not treat text reliance as a novice fallback to be designed out, nor assume a shared convention vocabulary between drawer and viewer. Finally, one participant refused AI assistance on ethical grounds, objecting that image models train on others' work—a reminder that "add generative AI" is not a neutral default, and one that mirrors the ethical questions the filmmaking community is raising about AI in creative workflows [76].

Whose criteria should govern? Drawers optimize what a moment means; evaluators recover clips through spatial composition, a keyframe prior, and a single distinctive anchor (Section 4.5). When author intent and audience recognizability diverge, a feedback-driven system must decide which leads. We propose routing authority by *provenance*: for an archival or third-party clip, weight the evaluator's criteria, since the goal is that someone who does not know the clip can identify it; for the author's own footage, weight the author's, since intent about meaning can legitimately override generic recognizability. This is a stance, not a resolution—the difference between authoring and recovering does not dissolve. What our data does settle is that the value of feedback lay in the adaptation it triggered rather than the score it produced: the most prevalent improvement mechanism was discovering devices *while evaluating* peers (33/41). Feedback systems in design and learning show the same asymmetry, where structured critique and rapid peer response drive revision more reliably than ratings [52, 63], and studies of reviewer feedback to scene-description authors find that specific, localized comments are what authors act on [71]. A feedback system's job is to manufacture that adaptation reliably, not merely to rate.

5.4 Design implications

We focus on two implications that follow directly from where our three sources agree, rather than enumerating guidance for every dimension we coded.

5.4.1 Give motion and order explicit devices, and verify that they are read. Camera movement is where the sketch fails, and it fails consistently across all three of our measures: it was absent from 23 of 42 sketch sets and almost never drawn directly (Section 4.2); it accounted for 10 of the 13 cases in which a drawer believed they had conveyed something the page did not carry (Section 4.3); and motion legibility is one of only two dimensions on which revision did not reliably help (0.60, interval including chance, Section 4.7). Order behaves differently but points the same way. An ordering arrangement is present in every set by construction, so no coder can record it as missing—yet order clarity is the weakest dimension in the pairwise comparison (0.59 [0.40, 0.76]). Presence of a structure is not communication of what it structures.

Research on static depiction of motion explains why adding a device is not sufficient. Cutting [21] surveys the solutions available to artists and scientists—dynamic balance, stroboscopic sequences, forward lean, photographic blur—and concludes that every satisfactory solution evokes motion while sacrificing the accuracy of the motion represented, so workable solutions are case-specific rather than general. Motion lines carry this tension further: they are not transparent depictions of a percept but learned graphical conventions whose comprehension varies with a reader's exposure to comics and manga [17, 38], and the same is true of sequential-image comprehension itself, which depends on fluency acquired through practice with a graphic system rather than being universal [15]. A drawer who adds an arrow has therefore not solved the problem; they have made a bet on a convention their particular evaluator may or may not read. That is exactly the bet our camera data shows them losing.

The implication is twofold. First, an authoring or generation system should supply *explicit relational and motion devices*—numbered or connected panels, directional and camera-path arrows, causal connectors—rather than relying on arrangement to imply sequence; arrows in particular are known to shift a diagram from structural to functional reading and are produced spontaneously when people are asked to convey process [42, 87], and schematic storyboarding has already demonstrated that outlines, arrows and text can carry motion in a single static image [33]. Text belongs in this set rather than beside it: our drawers used written labels to name what a frozen frame could not show, and adding text was the single most frequent revision operation (Section 4.6), which argues for treating language as a co-equal encoding channel rather than a caption laid over the real preview [41, 85]. Second, and because conventions are learned rather than universal, the system must *verify recovery rather than assume it*: supplying the device is only half of the design, and the verification half is what Implication 5.4.2 provides.

Why current technology does not solve this. A single-frame thumbnail has no sequence at all, and multi-frame generation can arrange panels without testing that a viewer recovers the intended order. For camera specifically the gap is documented: benchmarks built to evaluate whether models can understand and reproduce camera movement find this capacity largely uncharted [58, 95], and shot-attribute recognition remains a dedicated modelling problem rather than a solved input [79]. Neither selection nor generation currently has a motion vocabulary to offer an author, nor a way to check whether the one they produced was read.

5.4.2 Make recoverability by an unfamiliar viewer the objective, and test it in the loop. The reason guessing belongs in our paradigm is that it operationalizes differentiation: a preview must let a viewer who does not know the clip tell it apart from confusable alternatives. Our drawers pursued this objective explicitly, selecting for rarity over typicality

(Section 4.1) and running an informal generate-then-verify loop by imagining how a stranger would read a mark. Current pipelines do not. Keyframe-based multi-shot generation illustrates the point: STAGE [104] is motivated by the observation that keyframe-conditioned methods fail to maintain cross-shot consistency and to capture cinematic language, and addresses this with architectural memory and training schemes—that is, by improving the *plausibility and coherence of what is produced*, never by testing whether a viewer could identify the source from the result. Plausibility and recoverability are different objectives, and a competent, coherent, attractive preview can still fail to differentiate its source from its neighbours.

We therefore argue for a *differentiation-in-the-loop* stage: generate or select candidate previews, test their recognizability against a set of confusable decoys—an automated analogue of our guessing task—and revise toward the candidate that maximizes correct recovery, with text and annotation available as a repair channel when no single frame differentiates. Our own AI evaluation is a working instance of the test half of this loop (Section 4.7): given a clip summary and four candidate descriptions, judges preferred revised sketches on identifiability and overall recoverability, and did so consistently enough across three model families to serve as a practical, cheap proxy for a human decoder.

This objective is also the one that matters for accessibility, which was our original motivation. What happens, in what order, and why is both the information a viewer most needs when they cannot or will not watch, and the information our data shows is most likely to be lost. Work with blind and low-vision viewers finds that the descriptive content people need is precisely this event-level and relational information, and that what counts as sufficient varies with viewing context [45, 62]; work on multilingual captioning likewise argues that access is not served by a single linguistic channel [23]. Recoverability by an unfamiliar viewer gives these concerns a testable operational form: a representation is accessible only if someone without the source can recover its intended content.

Why current technology does not solve this. Selection optimizes engagement and aesthetics with no recognizability term; generation optimizes plausibility, not recoverability to *this* source; and text captions describe what is in a frame rather than the relations across frames, while remaining bound to one language. None of these pipelines models a decoder or runs a recovery test, so none can detect a plausible-but-undifferentiating result.

5.5 Limitations

5.5.1 Task format and the boundaries of the differentiation test. Participants encoded each clip into a prescribed four-panel sketch, and evaluators identified clips by forced choice among four candidates. Both constraints shape what our results can claim. The fixed panel budget determines what must be compressed and what is offloaded to text or arrangement, so our account of selection and channel use is relative to this format; a single frame, a longer sequence, or a short looping animation would redistribute that work. The closed candidate set matters more. Deployed previews compete against open collections in which far more clips are confusable, so our recoverability estimates should be read as optimistic, and the power of a single distinctive anchor will fall as the decoy set grows. Relatedly, our evaluators were fellow participants who shared the task context rather than fully naive viewers, so "recoverable" is measured against a partially informed decoder; a stricter protocol would likely widen the gaps we report rather than narrow them. Future work should vary the panel budget and scale the decoy set systematically, using naive viewers recruited outside the session, to establish how differentiation degrades as the competing set grows— the condition under which any deployed preview actually operates.

5.5.2 Stimuli, participants, and the conventions they bring. Our clips were curated film excerpts spanning a 2×2×2 motion and event design. The design is useful for reasoning about how clip type shapes encoding, but with one to three

coded sketches per cell any clip-type pattern is descriptive rather than inferential, and we treat it as such throughout. More consequentially, film is not the video most previews are made from: the priors our evaluators relied on—expecting the first panel to correspond to the opening, checking plausible spacing across the clip—presuppose well-formed edited material and may not transfer to vlogs, tutorials, sports or live streams. Our participants were also predominantly Chinese-speaking, and several conventions we observed, including typography-as-affect and warm/cool colour salience, are plausibly culturally situated; comprehension of sequential images and of motion lines is known to depend on exposure to particular graphic traditions [15, 38], so the specific repertoire we document should not be read as universal. Drawing skill and graphical literacy vary within our sample as well, and although picture-first and text-first dispositions were not reducible to skill, what can enter a sketch remains entangled with what a person can draw [56, 77]. Future work should replicate with user-generated video and with participants from different graphic traditions, testing whether the dimension-to-channel mapping we report is stable or whether the repertoire itself is what varies.

5.5.3 Coding, model judgment, and what the comparison can attribute. Our analysis rests on one authoritative coder's resolved coding of the sketches, consistent with guidance that reliability in qualitative HCI work is established through documented process rather than through a single agreement statistic in every case [67]; the scheme nonetheless remains a researcher construction, and the encoding levels are defined relative to it. Our model-based comparison inherits its own constraints. Large language models applied as judges are consistent and scalable but weaker on nuanced and context-dependent material than human evaluators [48], and although we balanced display position across repeats, a residual position tendency remained (A-minus-B = -0.200, 95% CI $[-0.378, -0.037]$), so we treat the comparison as an exploratory check that corroborates the descriptive account rather than as a controlled experiment. Most importantly, revision operations are descriptive classifications applied to whole sketch sets, and sets typically underwent several operations at once; the design therefore cannot attribute the observed improvement to any specific operation. Future work should manipulate operations directly— adding text alone, adding an explicit ordering device alone—and measure recovery against an unrevised control, which would convert our descriptive account of *what people change* into causal evidence about *what works*.

5.5.4 From characterizing a process to building and testing a system. We characterize how people create and repair previews and derive design guidance from it; we did not build or evaluate a preview system, so the implications in Section 5.4 are grounded hypotheses rather than validated claims. Two are directly testable. A *differentiation-in-the-loop* generator should be built and compared against engagement-optimized baselines, measuring whether candidate previews selected by recovery against confusable decoys are identified more accurately by unfamiliar viewers. And *explicit relational and motion devices*—numbered or connected panels, camera-path arrows, structured text channels—should be supplied in an authoring interface and tested on the dimensions that currently resist static encoding, namely camera, motion and order. Beyond these, the accessibility framing deserves its own evaluation: visual previews should be compared against text captions and frame-selection baselines on naive-viewer recovery of event, order and causal content, across language backgrounds, to establish whether a visual channel carries temporal information that language-bound captioning currently does not.

6 Conclusion

We studied how 42 people turned short film clips into static, four-panel visual previews, how peers read those previews, and how the sketchers revised them once a peer had tried to identify the source. Making a preview proved to be an act of strategic compression rather than neutral summarization: participants selected moments by weighing what carried

the event, what distinguished this clip from others, and what could be drawn clearly at all, and these three requirements frequently conflicted. To carry what depiction alone could not, they assembled a mixed notation of drawing, color, arrows, and written words, using words in particular to name actions that a frozen frame cannot show.

The translation into static form was selective. Entities and states — people, props, faces, poses — survived, while camera movement was the dimension most often left out and the one participants most often believed they had conveyed when the page did not carry it. Because sketchers held the clip and their own intention in mind, they filled in what a stranger could not, and had no way to notice the difference. Peer evaluation made that gap visible and actionable: revisions were dominated by *adding* information, and most often by adding words — the channel participants had treated as secondary. Comparing each original against its revision, revised previews were more identifiable and easier to reason about causally, while the temporal and dynamic content that had been hardest to encode in the first place remained roughly where it was.

Taken together, these results give the empirical grounding that visual captioning systems currently lack. They indicate that a preview should be judged by whether someone who has not seen the source can recover it, not by fidelity or predicted engagement; that relational content needs explicit, reader-recoverable devices rather than an arrangement that merely implies order; that text and annotation are a first-class encoding channel rather than a caption laid over the real preview; and that a feedback loop's value lies in the adaptation it triggers, not in the score it produces.

This paper contributes: (1) an empirical account of how people select moments from a video stream and encode them into static visual form; (2) a taxonomy of notation for dynamic film content, decomposed by channel and by information dimension; (3) a characterization of what resists visual encoding together with a characterization of the revision operations people perform in response to peer feedback; and (4) design implications for visual captioning techniques and for the human- and AI-in-the-loop systems that would improve them. Together, these show how the temporal content of video might be represented for accessible, language-independent understanding by building on how people themselves encode and refine it.

7 Appendix:

A Model comparison prompt

The template below was rendered once per sketch-set pair, substituting the film's ground-truth summary and its four candidate descriptions. It was issued to each model as a *single user-role message* containing the labelled SYSTEM and USER sections shown, rather than as separate native system and user messages; structured output was enforced through each provider's schema mechanism. Version A and Version B were the two attached four-panel sheets, with original/revised assignment balanced across repeats (§3.5.5).

SYSTEM

```
You are evaluating hand-drawn visual previews of short film clips. Each preview is a set of four sketched
panels made by a person who watched one clip and tried to make it identifiable to someone who has not seen
it. You will compare two versions of a preview for the SAME clip. Judge only how well each version
communicates the clip to a viewer who has never seen it. Ignore drawing skill, neatness, and artistic
quality except where they affect comprehension. Do not favor the longer or more detailed version by default.
Output valid JSON only, with no commentary.
```

USER

```
Clip summary (ground truth): {{ clip_summary }}
Candidate clip set the viewer must choose from:
{% for option in candidates %}- {{ option.letter }}: {{ option.description }}
{% endfor %}
```

```
Version A: the FIRST attached image, a four-panel sheet labelled Panel 1 through Panel 4.
Version B: the SECOND attached image, a four-panel sheet labelled Panel 1 through Panel 4.

For each dimension below, decide whether Version A or Version B better achieves it. You must choose A or B
for every dimension; do not output ties.

1. identifiability - From this preview alone, a viewer could pick the correct clip out of the candidate set
above.
2. event_clarity - A viewer could say what happens in the clip, not merely what objects or people appear.
3. order_clarity - A viewer could recover the order in which the depicted events occur.
4. causal_clarity - A viewer could recover why events happen, i.e. which event leads to which other event.
5. motion_legibility - A viewer could recover how subjects and the camera move.
6. overall - Overall, which version better lets an unfamiliar viewer recover this clip.

Return exactly:
{"identifiability":"A|B","event_clarity":"A|B","order_clarity":"A|B",
"causal_clarity":"A|B","motion_legibility":"A|B","overall":"A|B",
"rationale":{"identifiability":"<=25 words","event_clarity":"<=25 words",
"order_clarity":"<=25 words","causal_clarity":"<=25 words",
"motion_legibility":"<=25 words","overall":"<=25 words"}}
```

B Participant demographics

Table 3. Per-participant demographics and self-reported background (N = 42). Gender: W = woman, M = man. Language reports the languages participants said they were proficient in (Man = Mandarin, Eng = English, Yue = Cantonese, Nan = Hokkien). The four background items are self-reported on a 1–7 scale: *Draw* = how often the participant draws, sketches, or doodles; *Train* = formal training in drawing, art, or design; *Abil.* = self-rated drawing ability; *Write* = how often the participant writes to express ideas.

PID	Age	Gen.	Language	Draw	Train	Abil.	Write	PID	Age	Gen.	Language	Draw	Train	Abil.	Write
P1	26	W	Man	2	4	4	3	P22	24	W	Man	4	2	5	5
P2	24	W	Man/Eng	3	6	3	3	P23	20	M	Man/Eng/Yue/Nan	4	4	5	4
P3	20	W	Man/Eng	1	7	7	7	P24	22	W	Man	3	3	4	5
P4	24	W	Man/Yue	4	6	5	4	P25	23	W	Man/Eng	3	5	5	7
P5	26	M	Man	1	1	1	1	P26	27	W	Man/Eng	3	4	2	6
P6	30	W	Man	3	4	4	3	P27	21	W	Man	4	5	4	6
P7	21	M	Man/Eng	5	6	5	5	P28	20	W	Man	3	2	5	3
P8	23	W	Man	2	1	3	4	P29	21	W	Man	3	2	4	3
P9	24	W	Man	2	1	2	5	P30	21	W	Man	2	1	3	2
P10	21	W	Man/Eng	4	1	4	6	P31	22	W	Man	5	2	4	6
P11	20	W	Man/Eng	6	5	5	7	P32	25	W	Man/Eng	6	4	6	6
P12	22	W	Man	4	5	5	4	P33	21	M	Man	2	6	3	5
P13	23	W	Man	2	4	3	4	P34	23	W	Man	3	2	4	3
P14	20	W	Man/Eng	3	4	5	2	P35	23	M	Man	4	2	5	5
P15	22	M	Man	2	1	2	3	P36	18	W	Man	5	1	5	6
P16	22	W	Man	3	5	5	4	P37	24	W	Man/Eng	1	1	2	5
P17	20	W	Man	5	1	4	3	P38	21	W	Man	5	2	4	5
P18	20	M	Man	2	5	4	2	P39	18	W	Man	3	2	1	6
P19	21	W	Man	5	3	5	5	P40	18	W	Man	2	3	4	4
P20	21	W	Man/Eng	3	1	3	4	P41	29	W	Man/Eng	2	2	2	5
P21	20	W	Man	4	2	4	3	P42	29	W	Man	2	4	2	3

C Interview codebook

Prevalence is reported as the number of participants whose transcript contained the code (of 41 transcripts) and the number of coded segments. Codes are not mutually exclusive.

Table 4. Interview codebook. P = participants of 41; Seg. = coded segments.

Theme	Code	Definition	P	Seg.	Example
Choosing what to keep	Selection criterion	Stated reason for keeping or discarding a moment	39	164	Keeps a once-only car-flip over a recurring character
	Distinctive anchor	Building the set around one unambiguous object, and expecting a viewer to work outward from it	33	120	Anchors the set on a keychain and treats the rest as context
	Identify vs. explain	Distinguishing identifying the film from conveying a moment's meaning from pinning the segment	17	30	"identifies the shot but not the film's meaning"
	Rarity reasoning	Preferring content that occurs once over content that recurs, on distinguishability grounds	12	16	"appears many times … and lacks distinctiveness"
Encoding channels	Notation device	Use of color, arrows, marks, or written words as an encoding device	38	214	Adds color "mass" to limbs to make movement obvious
	Abstraction and density	Deliberate simplification, omission, or use of empty space	30	74	Draws people "simply … not so detailed"
	Picture-first vs. text-first	A stable disposition to lead with drawing or with words, independent of skill	25	53	A trained illustrator works text-first for "value per effort"
What resists encoding	Motion in a still	Subject or camera movement described as unable to survive a frozen frame	25	58	"from a single frozen frame you don't know what the hand is doing"
	Interiority	Emotion, reciprocity, or atmospheric stillness described as hard to show	20	54	A calm scene the participant "didn't know how to express, image or text"
	Off-frame content	Sound, voiceover, undepicted events, or empty landscape	12	29	Writes "sound of flowing water" rather than draw it
How evaluators identify	Spatial composition	Reading layout, position, depth, or camera angle as the first identification cue	25	66	"the spatial relationship is the primary clue"
	Timeline matching	Checking panels against the expected temporal spacing of the clip	24	59	Rules a candidate out if the first image is not early enough
	Evaluator simulation	Encoding by anticipating how a stranger will read a mark	19	30	Notes a stranger may not read a danger mark as intended
	Opening-frame prior	Expecting the first panel to correspond to the clip's opening	4	6	"the first frame is the film's opening"
Learning from feedback	Device adoption	Adopting a device after seeing a peer's sketch, often with regret at not using it earlier	33	71	"should have added text and color sooner"
	Named misreading	A specific element the evaluator read incorrectly, and the repair that followed	27	57	Drawn water read as "soil"; a railing read as "hurdles"
	Task misunderstanding	An incorrect model of the task that produced a fidelity error	5	10	Assumed the panels had to be chronological and mis-drew the ending
Tool and AI	Desired assistance	Where a tool could draft, render, select, or help comprehend	33	70	—
	Reservations	Ambivalence about automation, including refusal on ethical grounds	19	25	Declines AI help because models train on others' work

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